

David Dangerfield

Principal Software Engineer | Product, Platform, and Mission-Critical Systems

Consumer Products | Payments | Healthcare | Medical Devices | 20+ Years

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SUMMARY

Principal Software Engineer with over 20 years of experience designing, building, and securing enterprise platforms at scale across international consumer, payments, healthcare, and FDA-regulated software. Delivered modern web/app experiences and critical workflows for systems serving millions of users and supporting multi-billion-dollar revenue streams. Strong computer science foundation (MS, Purdue) and real-world experience spanning backend, frontend, test architecture, and systems engineering. Currently building and leveraging agentic AI development workflows to increase engineering velocity while maintaining essential human roles of expert analysis, direction, and review.

Technical Profile

Languages: TypeScript/JavaScript, Java, C++, C#, Python

Architecture & Paradigms: Microservices, Microfrontends, Event-Driven Architecture, Legacy Modernization, I18n

Backend & Frameworks: REST APIs, Spring Boot, Node.js, Message Queues, HIPAA-Compliance

Data Stores: PostgreSQL, MySQL, MongoDB, SQL Server, Redis, IBM DB2

Cloud & Infrastructure: AWS, Azure DevOps, Google Cloud Platform, GitHub Actions, Docker, Kubernetes

Testing, Tools, & AI: Copilot, Claude, Playwright, Cypress, Jest, xUnit

Frontend: Vue.js, React, Next.js, GraphQL, Angular, Design Systems, Accessibility, Component Libraries, Tailwind, eCharts

EXPERIENCE

Principal Software Engineer (Contractor)

Mar 2025 - Present

[Patterson Companies, Inc. \(Remote\)](#)

- Rewrote the bill pay system for dental and veterinary sales sites serving practices across North America, supporting a multi-billion dollar annual revenue stream with locale-aware code
- Translated Figma designs to TypeScript/Vue.js frontend extending the UI component library for consistency and integrating with .NET backend hosted in Azure and C# microservices to SAP
- Transitioned to agentic development using GitHub Copilot and Claude

Tech: Copilot, Claude, Node.js, TypeScript, Vue.js, Microfrontends, Microservices, C#, SQL Server, Azure DevOps, I18n

Lead Software Developer (Contractor)

Jul 2022 - Mar 2025

[Life Time Inc. \(Remote\)](#)

- Built web and app experiences for a major consumer fitness platform, driving hundreds of thousands of new accounts and app installs
- Replaced legacy libraries in core platform code for high-traffic consumer sites

Tech: GitHub Copilot, TypeScript, Vue.js, Bootstrap, Vite, Vitest, Playwright, Legacy Modernization, Mobile App

Senior Software Engineer (Concurrent Part-Time Contractor)

Feb 2018 - Mar 2024

[Nurocor - Digitalized Clinical Development \(Remote\)](#)

- Built a structured authoring platform for FDA clinical trial documentation, enabling compliant document templates, formal review workflows, and complete audit trails on every edit
- Implemented audit trail functionality ensuring full traceability, a core FDA requirement
- Platform recognized by the TransCelerate BioPharma ClinEco Protocol Review Challenge as Best Integrated Protocol Authoring and Review Solution

Tech: Kubernetes, HIPPA, Vue.js, Nuxt.js, Storybook, GraphQL, Component Libraries,

Principal Software Engineer Consultant (Contractor)

Jan 2021 - Jul 2022

[NextEra Analytics \(Remote\)](#)

- Built decision-support tools helping energy sector leaders direct billions of dollars in annual capital spending, replacing manual spreadsheet workflows with scalable web applications
- Developed executive dashboards and data visualizations translating complex capital allocation data into actionable business intelligence

Tech: TypeScript/JavaScript, Python, Terraform, AWS, Angular, Vue.js, GraphQL, eCharts, Storybook

Senior Software Consultant (Concurrent Part-Time Contractor) *Sep 2020 - Feb 2022*

Intricately (Acq. by HG Insights) (Remote)

- Partnered directly with the CTO to level up engineering practices across a globally distributed team
- Contributed to a B2B cloud spend intelligence platform subsequently acquired by HG Insights

Tech: TypeScript, Vue.js

Senior Software Engineer

Sep 2018 - Jan 2021

SmugMug (Remote)

- Developed and maintained SmugMug.com, one of the longest-running independent subscription-based web services, hosting billions of photos for millions of customers worldwide
- Replaced legacy custom payment processing with Stripe integration, modernizing financial infrastructure for a high-volume consumer platform

Tech: AWS, Node, JavaScript, React, Tailwind, GraphQL, Continuous Deployment, Legacy Modernization, Accessibility

Advisory Software Engineer, Watson Health Consulting Services *Nov 2015 - Aug 2018*

IBM Services Group (Rochester, MN)

- Developed early AI-powered pharmaceutical research experiences using IBM Watson NLP to surface relationships across biomedical literature
- Built clinical decision support applications for healthcare workers including diabetes treatment recommendations and patient history visualization, under HIPAA compliance protocols
- Delivered cognitive AI applications for financial clients including a top 50 US bank
- Co-founded FED@IBM Rochester, an internal frontend developer community of 30+ engineers hosting twice-monthly sessions on JavaScript, accessibility, UI automation testing, and tooling

Tech: IBM Bluemix (Cloud), Vue.js, Angular, React, JavaScript, I18n, D3.js, Mobile Web App, Legacy Modernization

Senior Web Application Developer

Aug 2013 - Oct 2015

Preventice Solutions (now Boston Scientific) (Rochester, MN)

- Built frontend and backend systems supporting BodyGuardian, an FDA-cleared wearable cardiac monitoring device transmitting patient data to clinical care teams
- Developed the Notification Service that delivered cardiac alerts to clinical staff, a critical component in a patient-care pipeline requiring high reliability
- Shipped within FDA and HIPAA constraints, translating strict compliance requirements into reliable software delivery

Tech: Java, C#, REST APIs, ActiveMQ, Identity Management, SAML SSO, ASP.NET, SQL Server, Angular

Staff Software Engineer, Team Lead, Test Architect

May 2002 - Aug 2013

IBM Systems Group (Rochester, MN)

- Ported IBM Directory Server (LDAP) to IBM i, a commercially shipped enterprise product.
- Architected test infrastructure and led functional testing across international teams
- Automated software update packaging, eliminating 500+ person-hours of manual effort annually and enabling self-service builds across teams

Tech: C++, Identity Management & Security, REST APIs, Java, JavaScript, Perl, IBM Db2, I18n

Software Engineering Intern

May 2001 - Aug 2001

Intel Corp. (Hillsboro, OR)

- Developed face-based search applications for the consumer photo market

Tech: Java, Image Processing Integration

EDUCATION

Master of Science in Computer Science | Purdue University, 2002

- Theory of Computational, Advanced Networking, Distributed Databases, OS Implementation, Cryptography

Bachelor of Science in Computer Science | Brigham Young University, 2000

- Research assistant to Dr. Tony Martinez developing neural network training model for handwritten character recognition in C